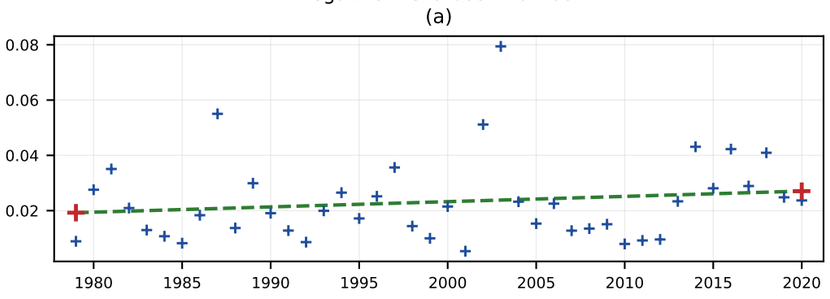
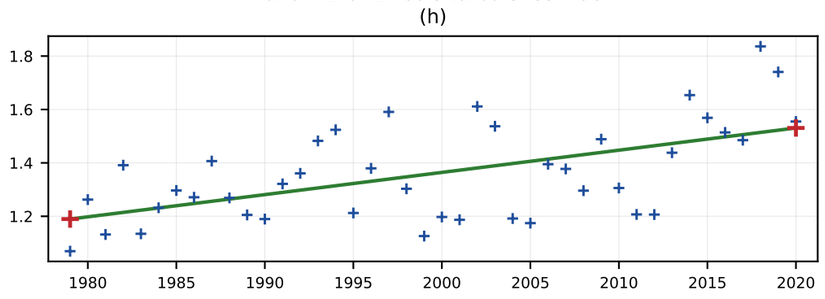
Prosser (2023) Figure 4 — his layout, his titles, drawn on OUR y axes
blue crosses: our 42 annual values · red crosses: our fitted 1979 and 2020 · solid green: significant at p=0.05
under each panel: his printed rel/abs/p, then ours (rel = % change 1979→2020, abs = slope in %/yr)

Negative Richardson number
(a)



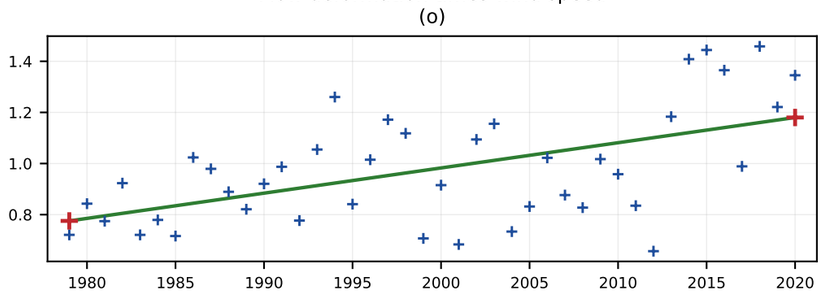
his rel=-7.5 abs=-0.000 p=0.6
ours rel=+40.6 abs=+0.000 p=0.32

Variant 2 of Ellrod's turbulence index
(h)



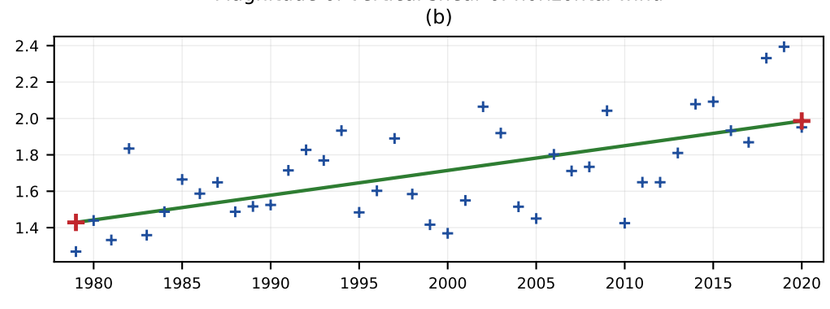
his rel=+27.0 abs=+0.009 p=3e-5
ours rel=+28.7 abs=+0.008 p=1e-4

Flow deformation times wind speed
(o)



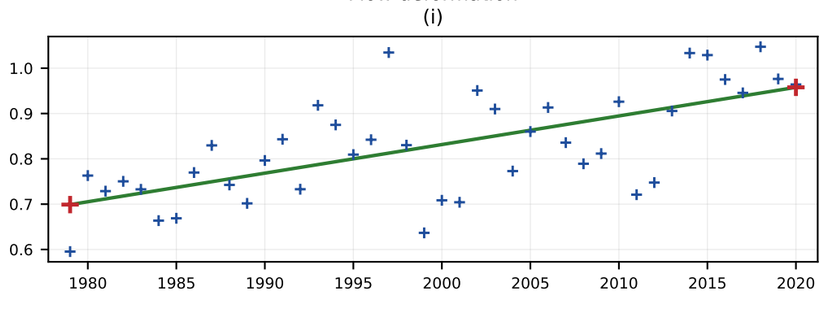
his rel=+50.9 abs=+0.010 p=3e-4
ours rel=+52.2 abs=+0.010 p=2e-4

Magnitude of vertical shear of horizontal wind
(b)



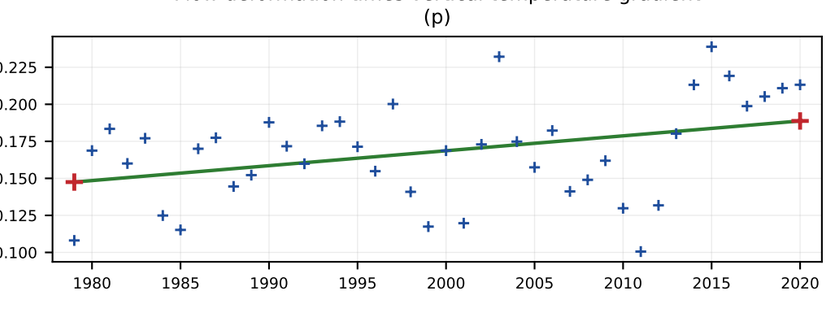
his rel=+38.4 abs=+0.016 p=8e-7
ours rel=+39.0 abs=+0.014 p=9e-6

Flow deformation
(i)



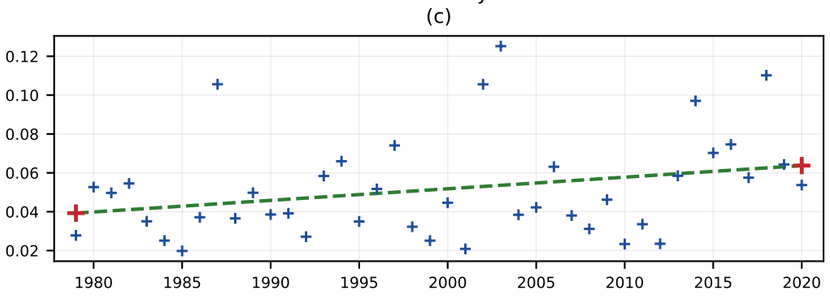
his rel=+37.5 abs=+0.006 p=6e-6
ours rel=+37.0 abs=+0.006 p=2e-6

Flow deformation times vertical temperature gradient
(p)



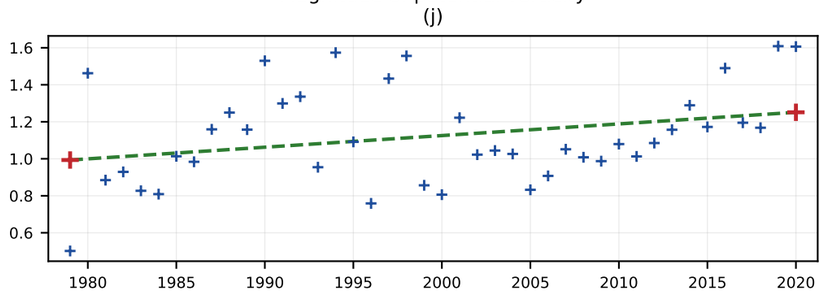
his rel=+33.8 abs=+0.002 p=1e-3
ours rel=+28.0 abs=+0.001 p=0.017

Colson-Panofsky index
(c)



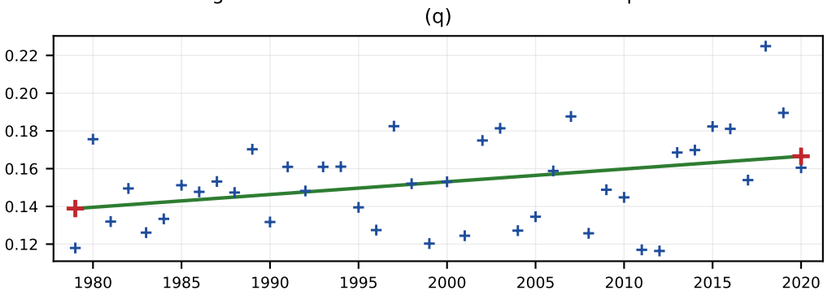
his rel=+45.4 abs=+0.003 p=2e-3
ours rel=+62.3 abs=+0.001 p=0.071

Magnitude of potential vorticity
(j)



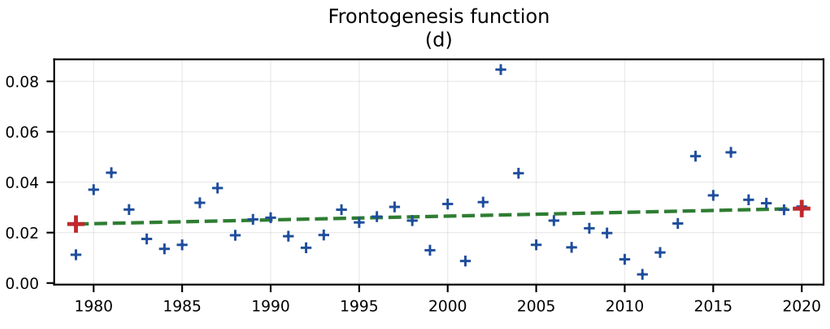
his rel=+50.6 abs=+0.020 p=4e-7
ours rel=+26.0 abs=+0.006 p=0.055

Magnitude of residual of nonlinear balance equation
(q)



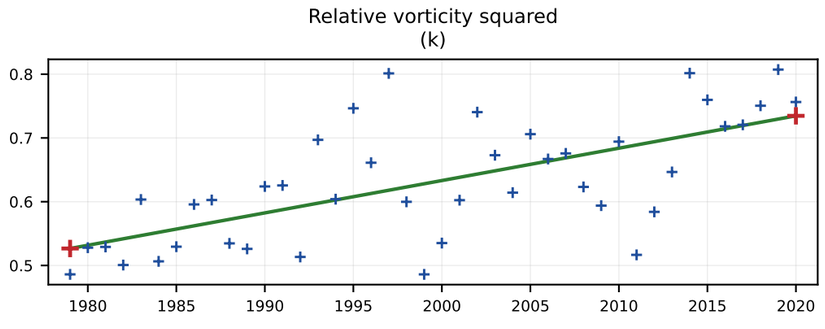
his rel=+31.6 abs=+0.006 p=6e-5
ours rel=+20.0 abs=+0.001 p=0.025

Frontogenesis function
(d)



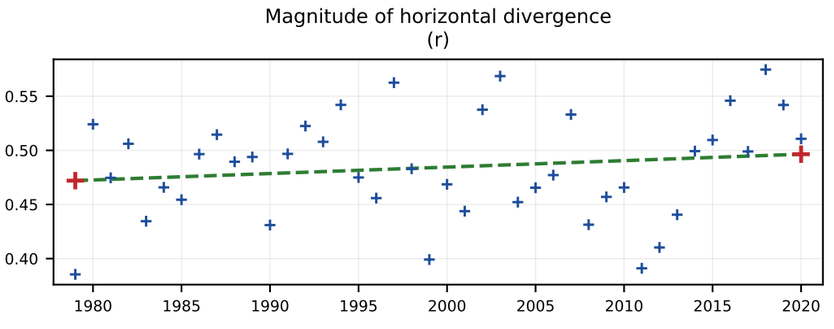
his rel=+0.3 abs=+0.000 p=1
ours rel=+26.2 abs=+0.000 p=0.42

Relative vorticity squared
(k)



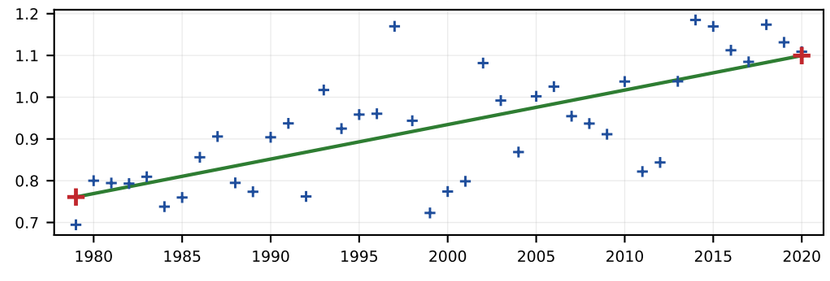
his rel=+37.1 abs=+0.004 p=2e-5
ours rel=+39.5 abs=+0.005 p=2e-6

Magnitude of horizontal divergence
(r)



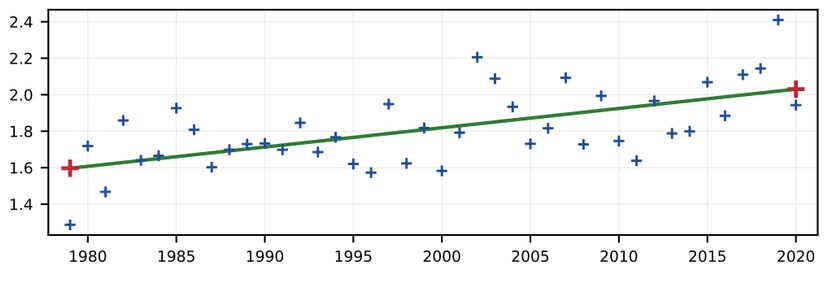
his rel=+6.7 abs=+0.001 p=0.2
ours rel=+5.2 abs=+0.001 p=0.33

Brown index
(e)



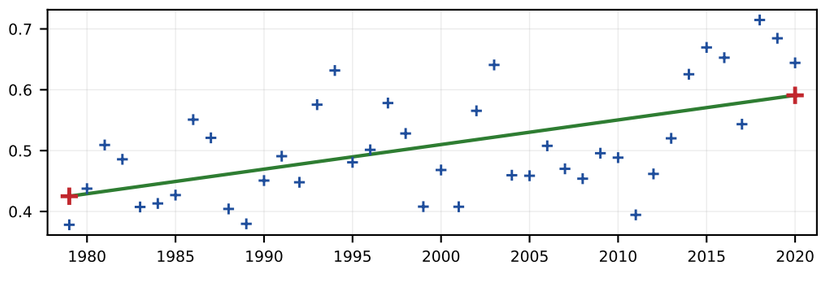
his rel=+45.4 abs=+0.008 p=2e-7
ours rel=+44.5 abs=+0.008 p=9e-8

Magnitude of horizontal temperature gradient
(l)



his rel=+26.0 abs=+0.011 p=2e-5
ours rel=+27.1 abs=+0.011 p=2e-5

Version 1 of North Carolina State University Index
(s)



his rel=+4.3 abs=+0.000 p=0.7
ours rel=+39.0 abs=+0.004 p=2e-4

Brown energy dissipation rate
(f)

his rel=+46.2 abs=+0.018 p=5e-8
ours rel=+46.1 abs=+0.016 p=5e-7

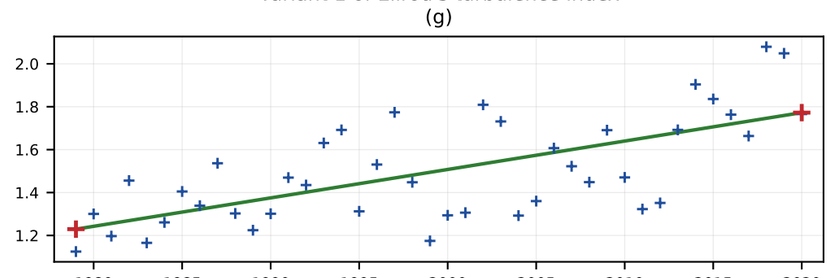
Wind speed
(m)

his rel=+75.6 abs=+0.005 p=0.02
ours rel=+73.9 abs=+0.005 p=0.014

Negative absolute vorticity advection
(t)

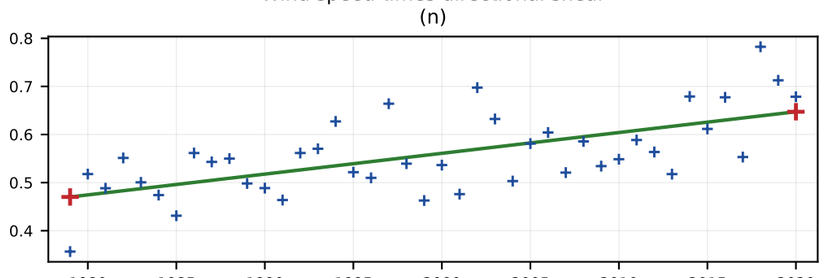
his rel=+28.4 abs=+0.005 p=1e-3
ours rel=+28.8 abs=+0.005 p=1e-3

Variant 1 of Ellrod's turbulence index
(g)



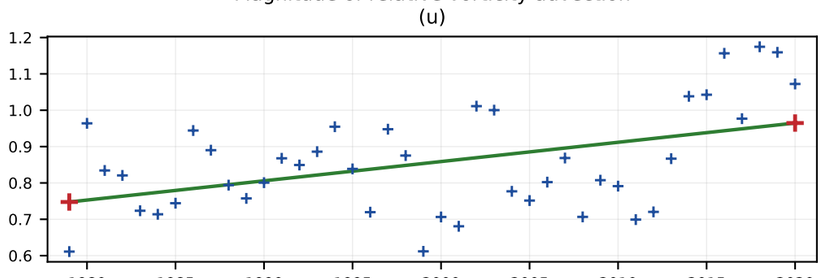
his rel=+42.3 abs=+0.014 p=3e-7
ours rel=+44.2 abs=+0.013 p=1e-6

Wind speed times directional shear
(n)



his rel=+27.7 abs=+0.006 p=2e-5
ours rel=+37.7 abs=+0.004 p=6e-6

Magnitude of relative vorticity advection
(u)



his rel=+29.0 abs=+0.006 p=3e-3
ours rel=+29.1 abs=+0.005 p=3e-3